SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 20 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I P V GIRIDHAR / 1972260022 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?

   Ans: Data Science can predict at-risk students by analyzing past data such as attendance, internal marks, assignment scores and study habits. After the data is collected and cleaned, EDA reveals which factors are linked to failure. A classification model (e.g., logistic regression or random forest) is then trained and tested to predict who is likely to fail. The college can identify these students early and provide counselling, remedial classes and extra support, which improves results and reduces failures through data-driven decisions.

2. A hospital has patient records for many years. How can Data Science be applied?

   Ans: Data Science can turn years of patient records into useful medical insight. The data is first cleaned (missing values, duplicates) and explored to find patterns. Machine learning models can then predict diseases and their risk, helping in prevention. Medical images such as X-rays and MRI can be analyzed to support diagnosis, and treatment can be personalized to each patient's history. This improves accuracy of care, reduces cost and helps doctors make better decisions.

3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

   Ans: This is the Personalization use of Data Science, implemented through a recommendation system. The website collects customer data such as browsing history, searches, past purchases and ratings. Techniques like collaborative filtering find customers with similar behaviour and suggest products they are likely to buy, as Amazon does. This gives each customer a tailored experience, increases sales and improves customer satisfaction and loyalty.

4. A bank wants to detect suspicious transactions. How can Data Science be useful?

   Ans: Data Science supports fraud and risk detection in banking. Historical transaction data (amount, location, time, frequency) is used to learn the normal behaviour of each customer. Anomaly detection and classification models then flag unusual transactions, such as a sudden large payment from a new location, in real time. The bank can block or verify such transactions before the fraud happens, which reduces financial losses and increases customer trust.

5. A company wants to reduce customer loss. How can Data Science support this?

   Ans: Data Science helps through churn prediction, a predictive modelling task. Customer data from the CRM database, usage history and feedback is collected and cleaned. A classification model is trained to identify customers who are likely to leave, and EDA shows the reasons, such as poor service or high price. The company can then take targeted retention actions like offers, better support or personalized communication. This reduces customer loss, with success measured by the drop in churn rate.

6. A school wants to improve student performance. Mention one benefit of Data Science in this case.

Ans: One major benefit is better decision-making. Data Science converts raw student data such as marks, attendance and test scores into actionable insights. It shows which subjects, topics or student groups are weak, so the school can plan extra classes, change teaching methods and give personalized support. Decisions are therefore based on facts and data instead of intuition, which leads to improved student performance.

7. A supermarket wants to know which products sell more during festivals. How can Data Science help?

   Ans: Data Science helps through demand forecasting, which is predicting future trends. Past festival-season sales data is cleaned and analyzed to find which products sell more and in what quantities. Machine learning models forecast the demand for the coming festival. The supermarket can then plan stock and inventory, avoid shortages or overstocking, arrange the supply chain in advance and offer targeted discounts. This increases sales and profit and reduces waste.

8. A transport department wants to reduce traffic congestion. How can Data Science be applied?

   Ans: Data Science can be applied through traffic pattern analysis and route optimization. Data from GPS devices, road sensors and cameras (often streaming data) is analyzed to find congestion-prone areas and peak hours. Predictive models forecast traffic and suggest alternate routes, and signal timings can be adjusted accordingly. This is also part of smart city planning. It reduces congestion, travel time and fuel use and improves public transport planning.

9. A farmer wants to improve crop yield. Give one use of Data Science.

   Ans: Data Science can predict crop yield and optimize farming decisions. Data such as soil quality, rainfall, temperature, humidity and past yields is collected from sensors and weather sources. Predictive models analyze this data to recommend the best crop, sowing time, and the right amount of water and fertilizer. The farmer can also detect pests or disease early. This increases yield, reduces cost and avoids wastage of resources.

10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?

    Ans: The data shows a combination of several facets of data. The Excel tables are structured data, arranged in rows and columns and stored in a database. The customer reviews are natural language data, a special type of unstructured data with no fixed format. The images and videos fall under the audio, video and image facet, which is also unstructured. Hence the company is dealing with structured as well as unstructured data.

11. A weather app updates temperature every minute. Which facet of data is involved?

    Ans: The facet involved is streaming data. It is generated continuously by many sources and sent in small records (kilobytes), here the temperature readings sent every minute by weather sensors. Other examples are log files, social network feeds and telemetry from connected devices. Since this data is created automatically without human entry, it is also machine-generated data, and it needs real-time processing tools such as Kafka, Spark or Flink.

12. A university stores student details in rows and columns. What type of data structure is this?

    Ans: This is structured data. It is arranged in a fixed format of rows and columns, where each row is a record and each column an attribute (e.g., name, register number, department). It is stored in a database management system such as MySQL or PostgreSQL, and can be retrieved and processed easily using SQL queries. An Excel table is another common example of structured data.

13. Customer comments on social media are collected for analysis. What type of data is this?

    Ans: This is unstructured data, specifically natural language (text) data. Social media comments do not follow a fixed row-and-column format, so they are difficult to store and retrieve in a traditional database. Natural Language Processing (NLP) is used to understand the meaning of the text, for example in sentiment analysis, to find whether customers feel positive, negative or neutral about a product or service.

14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?

    Ans: The Big Data characteristic applied is Volume, which refers to the massive size of data being generated. A mobile app produces huge amounts of daily user activity data (clicks, logs, sessions), which is too large for traditional tools like Excel. It needs distributed storage such as HDFS and NoSQL databases. Velocity (speed of generation) and Variety also apply, but the main characteristic here is Volume.

15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?

    Ans: The data quality issues present are incomplete data (missing age values) and inaccurate or incorrect data (wrong phone numbers). Poor quality data gives wrong results and models, so it must be fixed through data cleansing. This includes handling missing values by filling them with the mean or median or removing the rows, correcting errors, standardizing formats and removing duplicates. This relates to the Big Data characteristic Veracity, meaning trustworthiness of data.

16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?

    Ans: The important characteristic is Value. Value means the business insights and benefits derived from data. Collecting a large amount of customer data is useful only when it is analyzed to find buying patterns, preferences and trends that support sales decisions, such as targeting customers, pricing and stock planning. Data without value gives no benefit, so converting data into actionable decisions is the real purpose of Big Data.

17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?

    Ans: The Data Storage component of the Big Data ecosystem is used. HDFS (Hadoop Distributed File System) stores very large files by splitting them and distributing them across multiple machines in a cluster. It is reliable and scalable because the data is

replicated, so it is not lost if one machine fails. Other storage systems in the ecosystem are NoSQL databases such as HBase, Cassandra and MongoDB.

18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?

    Ans: Apache Spark can be used for this. It processes large datasets in parallel across a cluster using fast in-memory computation, and supports both batch and real-time processing. Another option is Apache Hadoop MapReduce, a batch processing framework that divides the job into smaller tasks, runs them in parallel on different machines and combines the results. For stream processing, Apache Flink can be used. These tools belong to the Data Processing component.

19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?

    Ans: The suitable component is Data Ingestion, which collects and imports data into the ecosystem. Apache Kafka, a distributed streaming platform, is best for collecting live data from IoT sensors because it handles a continuous stream of data from many sources in real time. Apache Flume can be used for log data and Sqoop for transferring data from relational databases to Hadoop. The ingested data can then be processed by Spark or Flink.

20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

    Ans: Data visualization and analytics tools such as Tableau or Power BI can be used. They convert Big Data results into dashboards, charts, graphs and reports that decision-makers can easily understand without technical knowledge. For real-time search and visualization of data, Elasticsearch with Kibana can be used. Visualization shows trends, patterns and key findings clearly, so managers can make quick, data-driven decisions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand